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KneeTrack Pro

PATELLA TRACKING KNEE BRACE



Design Proposal for Wearable Device using a Time-of-Flight Sensor to Quantify Mediolateral Patella Motion

Meet the Team



Jack Fairhurst
Software Engineer



Dan Kekeh
Team Lead



Khaled Gadalla
CAD Engineer



Emma Graham
Ombudswoman



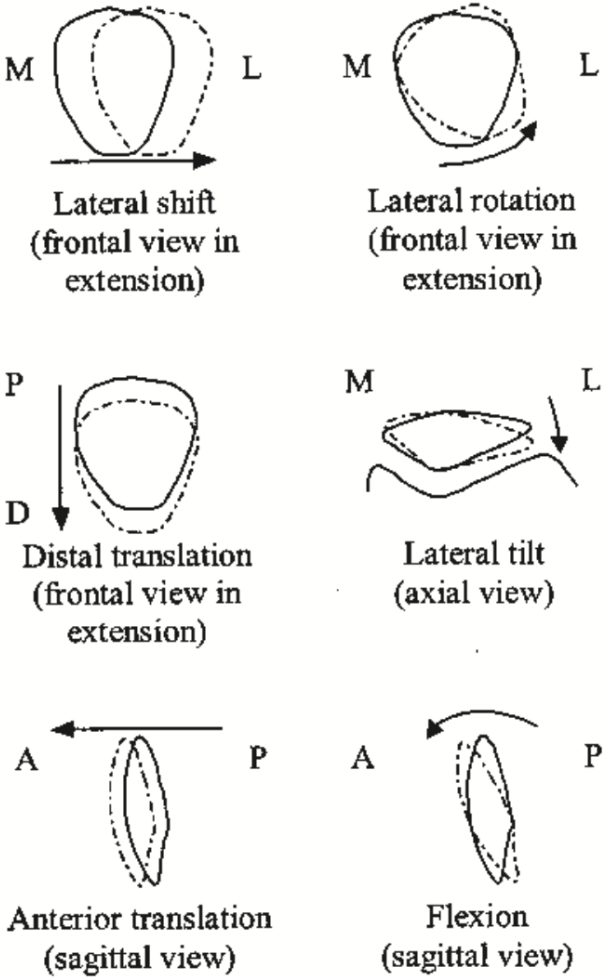
Joshua Lipshaw
Hardware Engineer



Zixuan Zhang
Material Testing Engineer

Clinical Problem

- **Patella Maltracking** – Deviation of the patella from normal motion (1)
- Affects **23%** of the population (2)
- Leads to **severe pain** (2)
- Current diagnosis does not include **quantitative** measurements (3)



1. Garza-Borjón et al. 2024. *Orthopaedic Reviews.*
2. Smith et al. 2018. *PLoS One.*
3. Jibri et al. 2019. *Insights Imaging.*

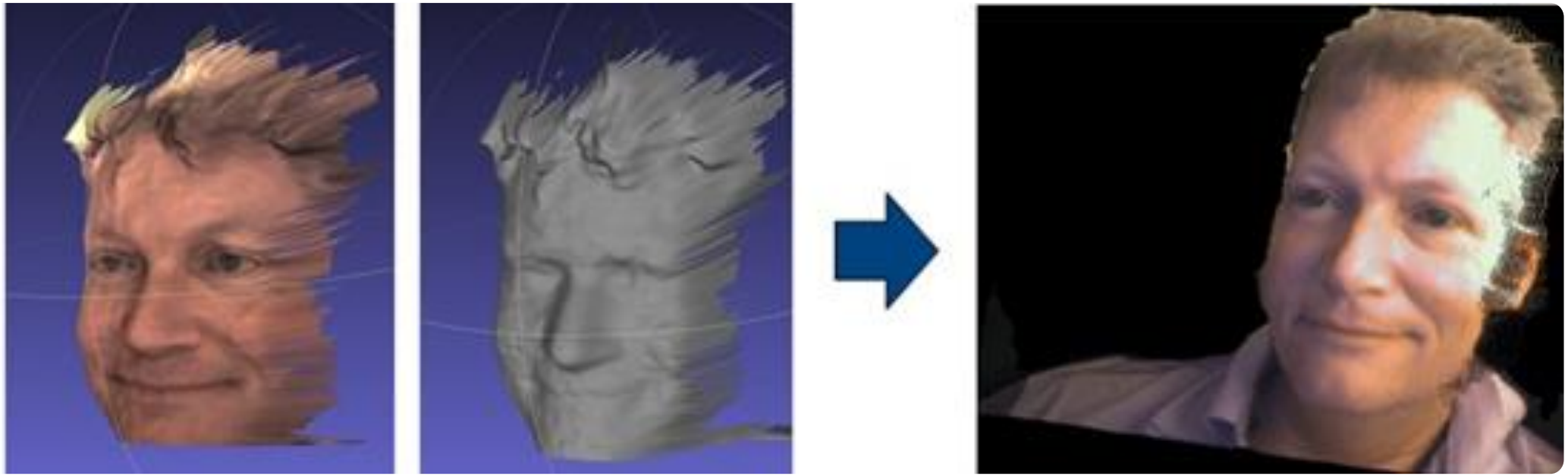
Device Aim

- Quantify mediolateral patella motion to **assist** clinicians in making maltracking **diagnoses**
- Increases speed of assessment, helping the patient **out of pain** more quickly.
- **Class 1m** – non-invasive, measuring device (1)



1. NHS UK. 2025. *Understanding UK MDR 2002 Regulations for Medical Devices.*

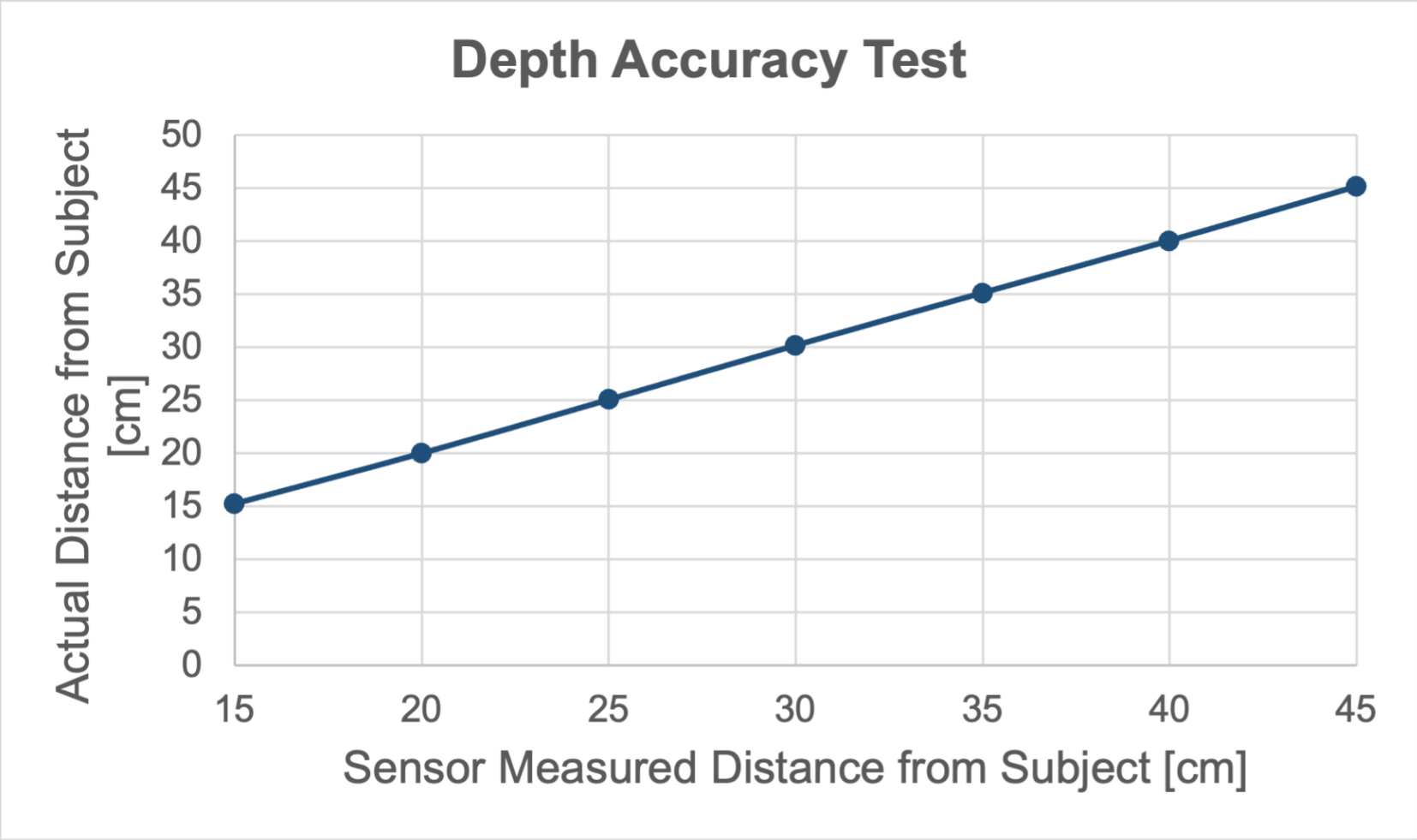
Depth Tracking



- Uses **time-of-flight sensors** in iPhone 14 Pro Lidar Scanner
- Sensors capture a 3D point cloud with RGB data
- Allows precise coordinate tracking of points of interest, with **minimal skin artefact**

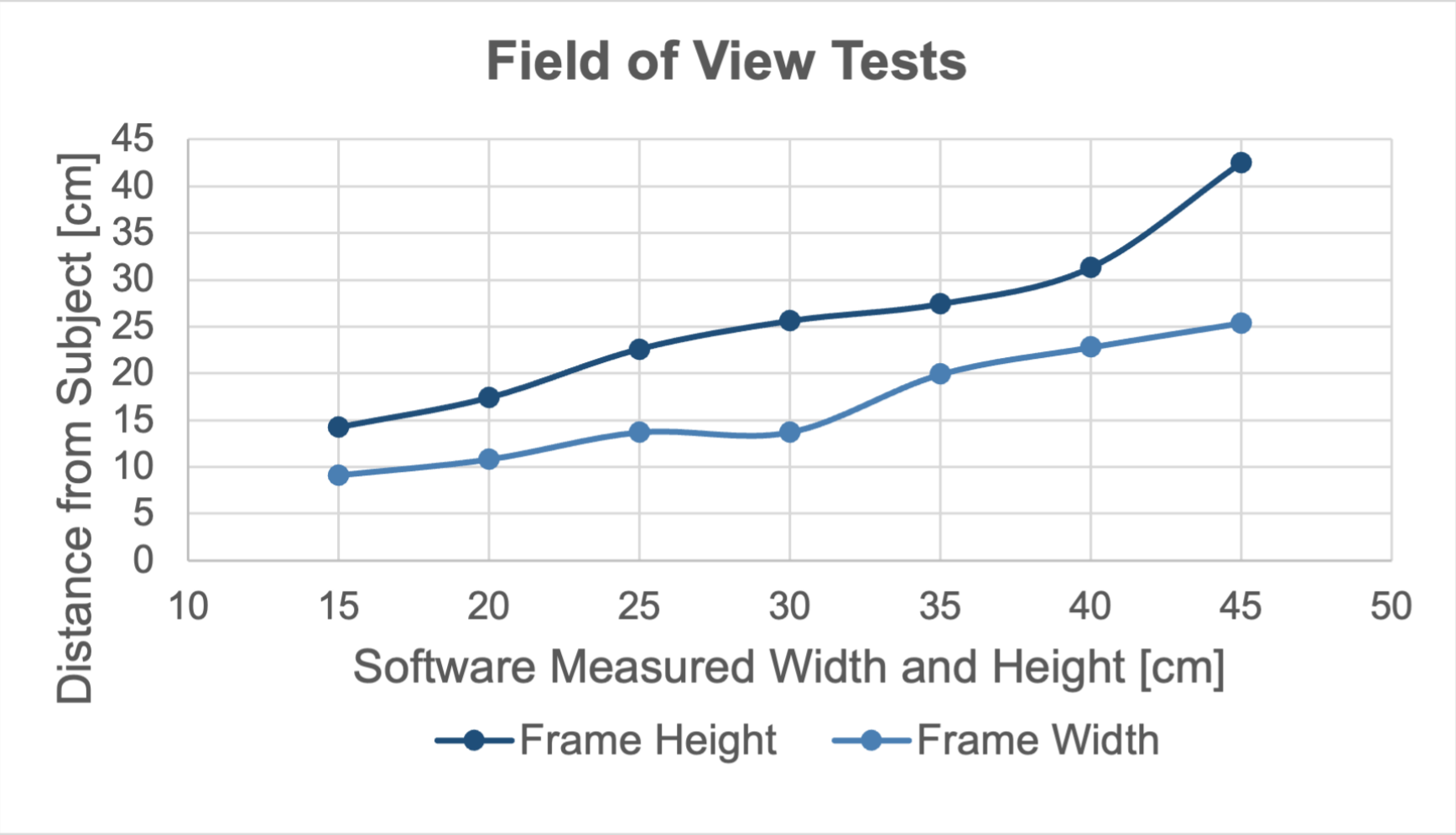
Images: Intel®

Depth Determination



Accuracy: ±1.013mm

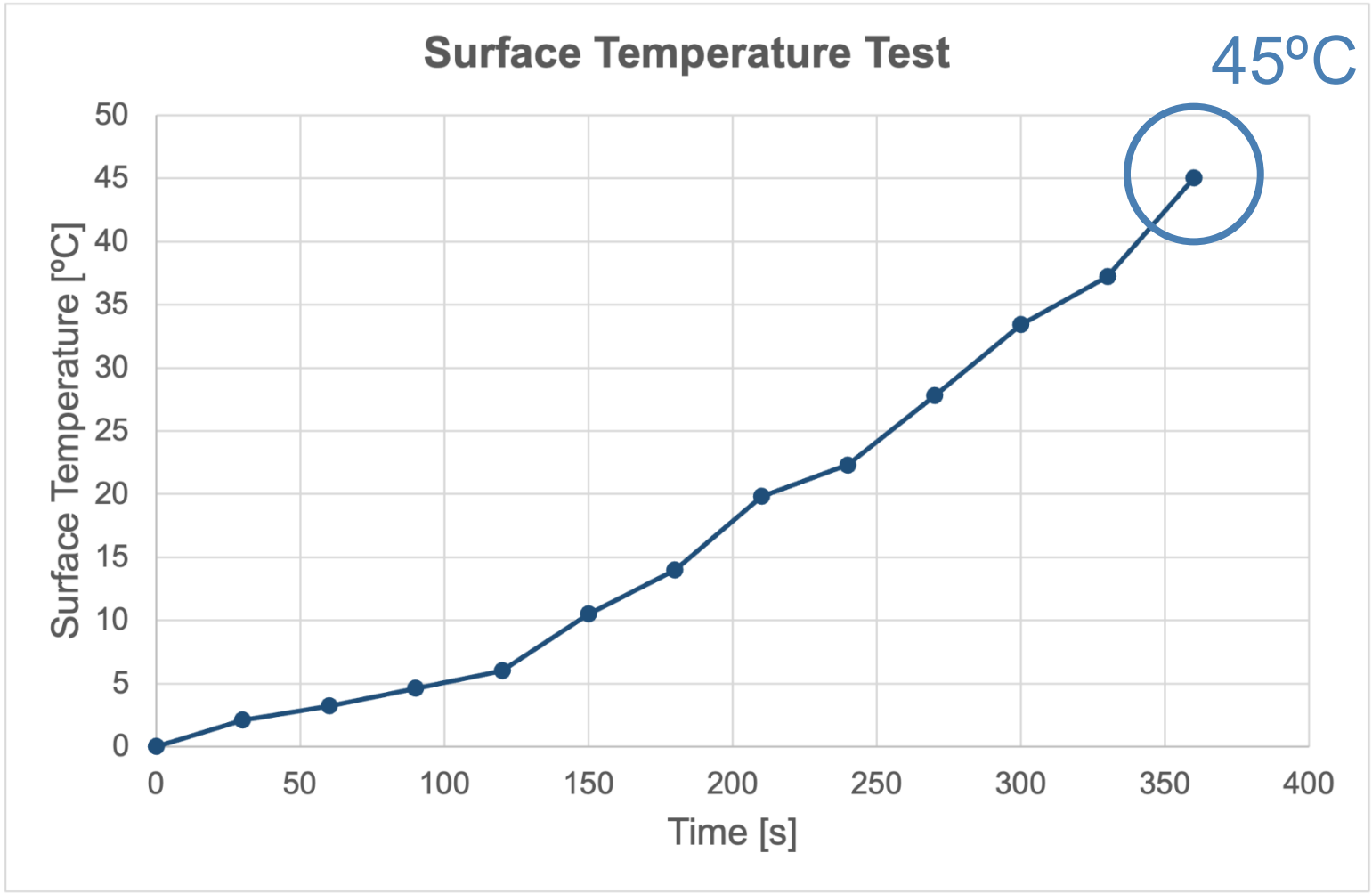
Depth Determination



1. Frings et al. 2020. *Scientific Reports*

Chosen 15cm distance has a 9.1cm frame width, covering observed 1.93cm range in literature (1)

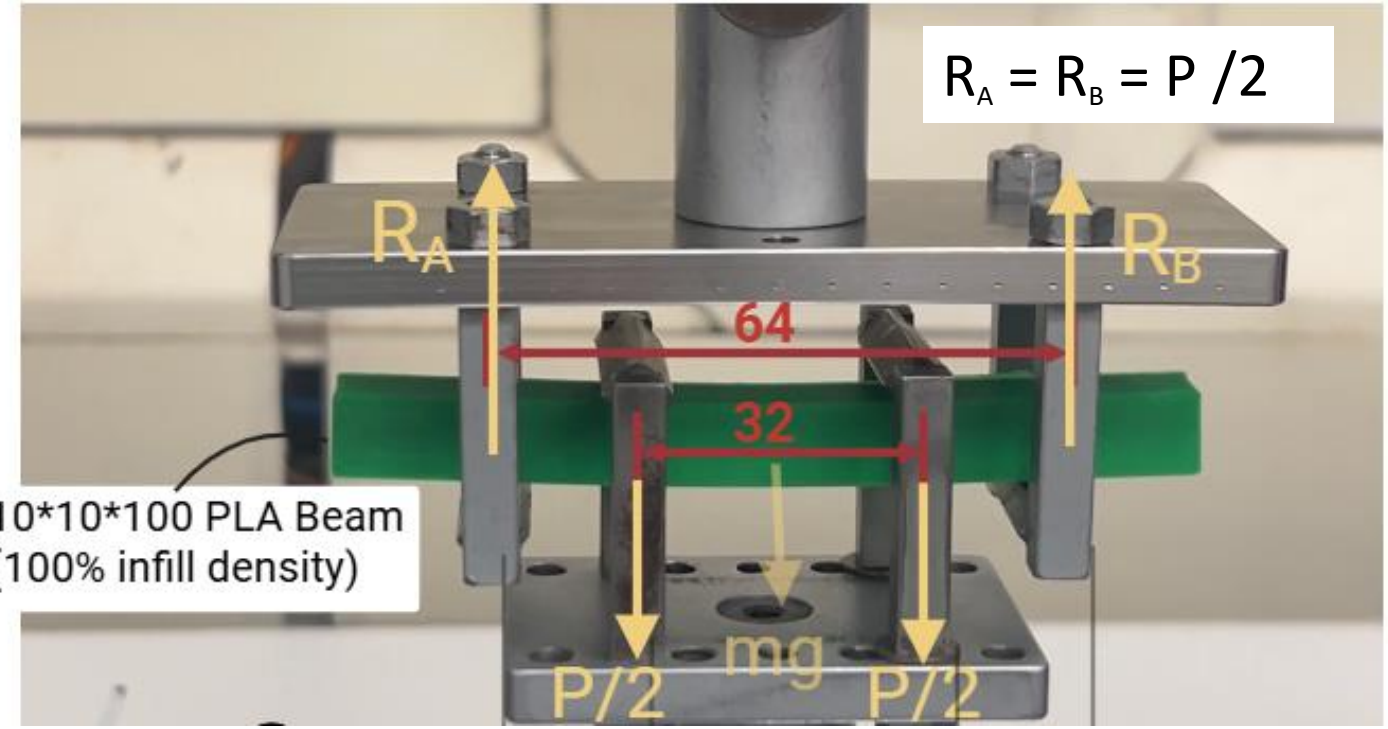
Battery Testing

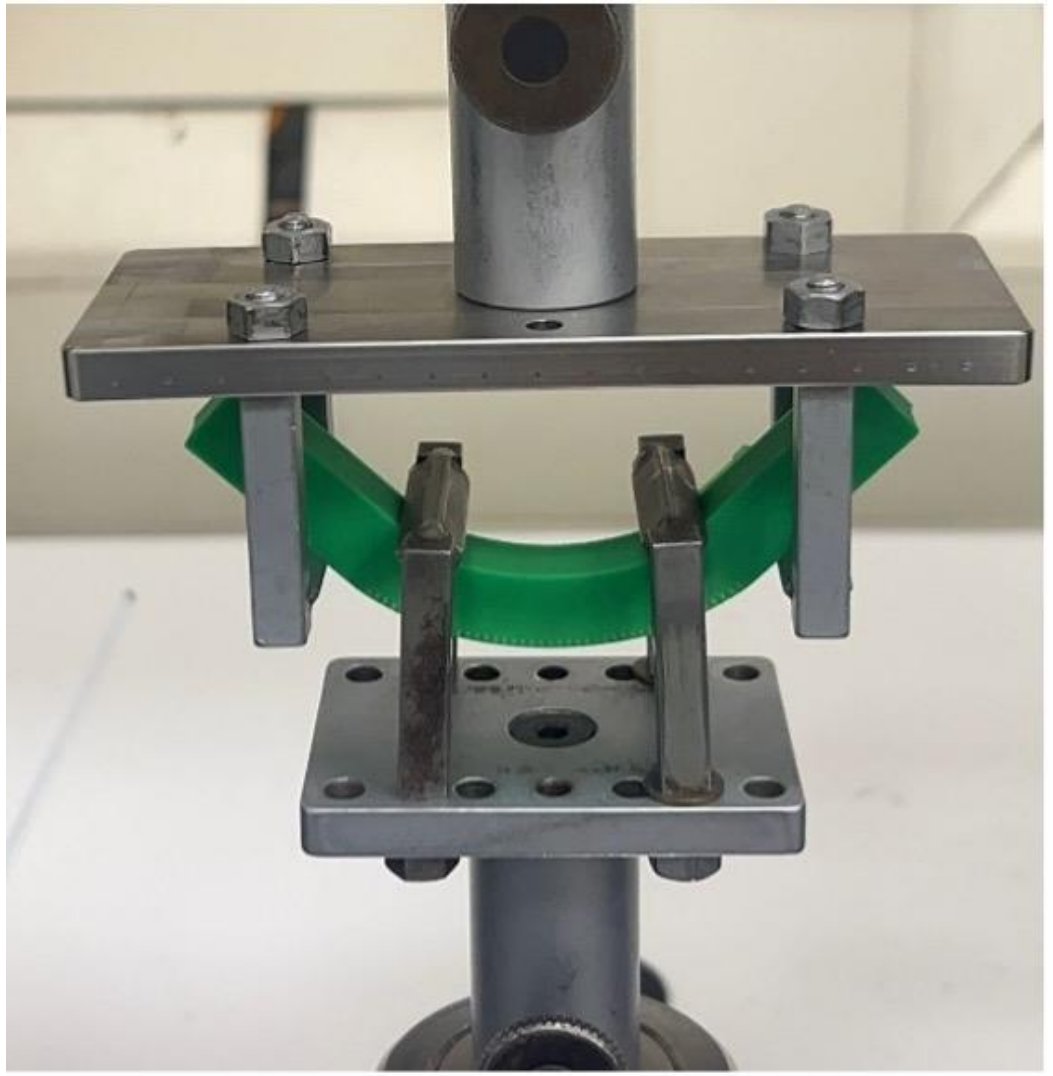


1. Geng et al. 2006. *Annals of Occupational Hygiene*



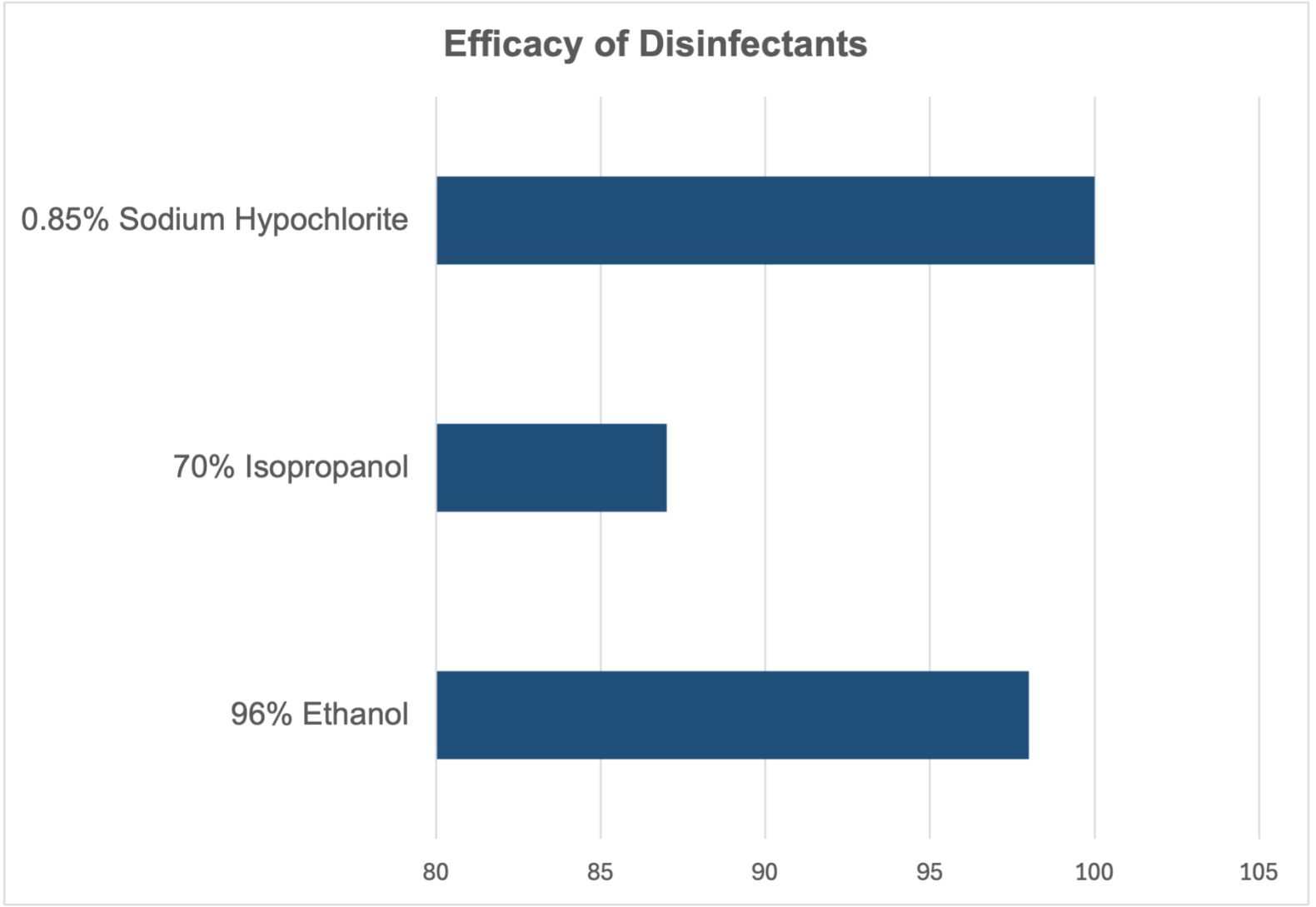
Material Testing





- Four-point bending test found flexural yield stress to be **81.9MPa**
- Free body diagram calculations predict maximum operating stress of **0.136MPa**
- Safety factor of **10**

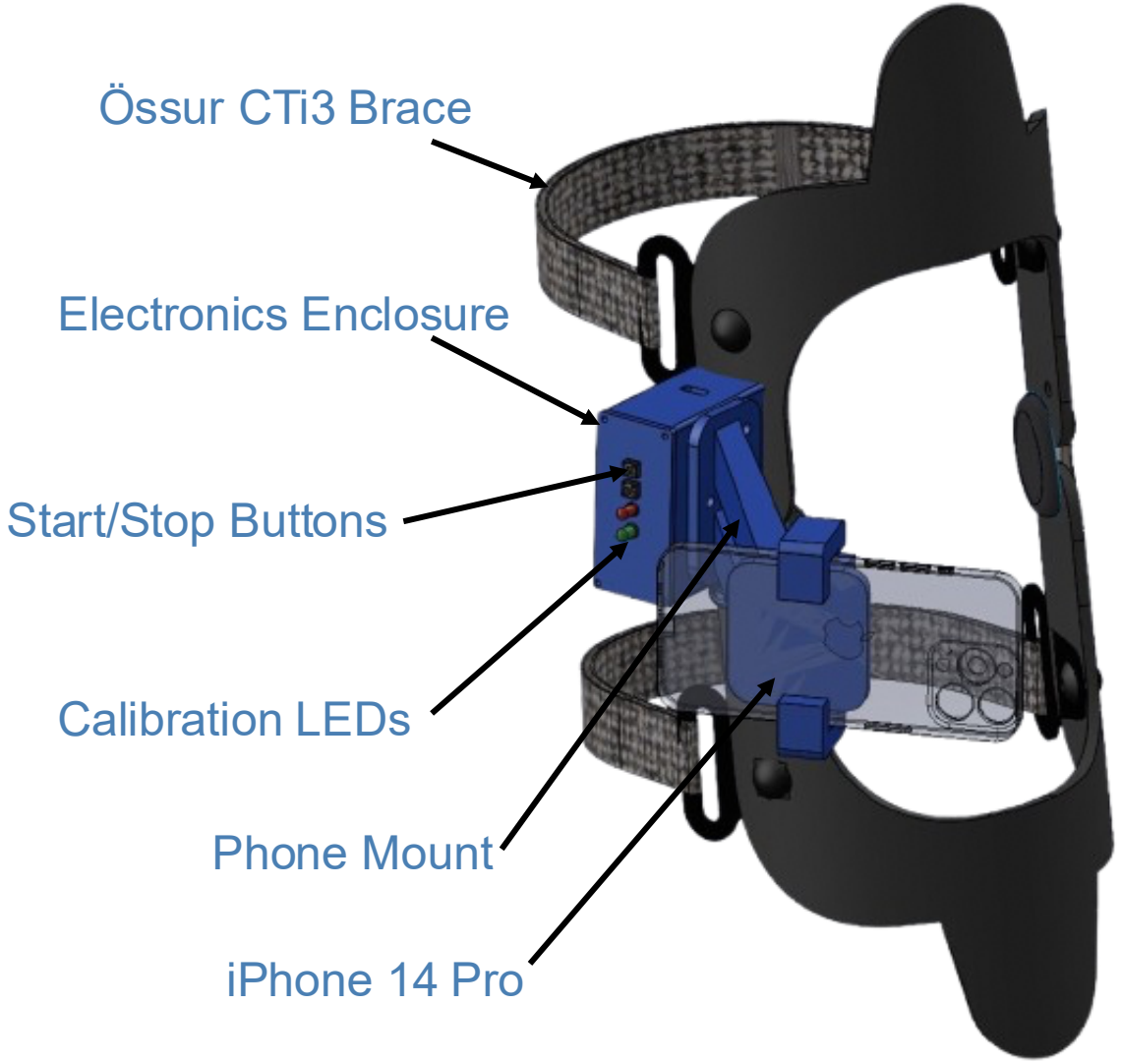
Material Disinfecting



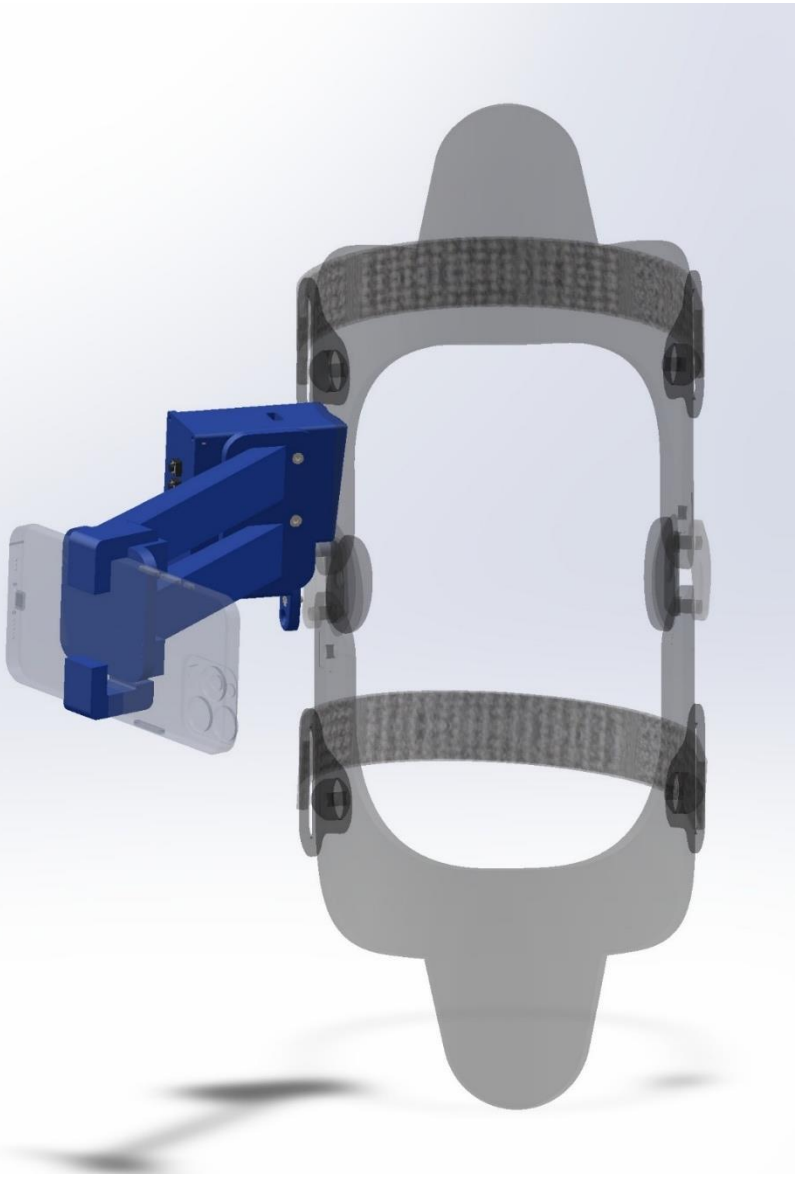
1. Vaňková et al. 2020. *PeerJ.*

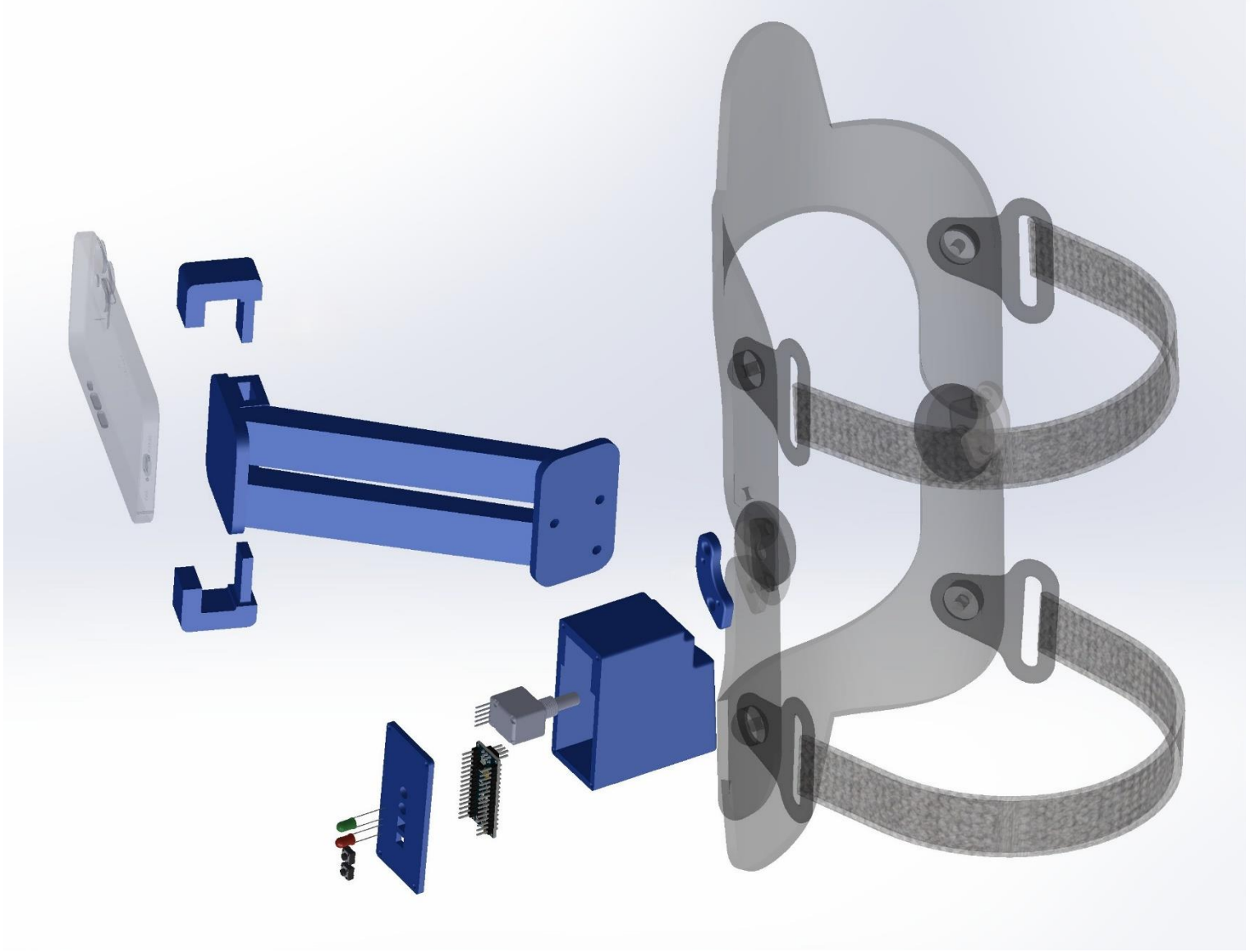
CAD Design

- **Bespoke Hinge Cap** – to preserve polycentric hinge motion
- **Rotary Encoder** – embedded in hinge
- **Functionality** – as knee flexes, encoder shaft rotates
- **Output –** hinge vs knee flexion angle



Exploded CAD View



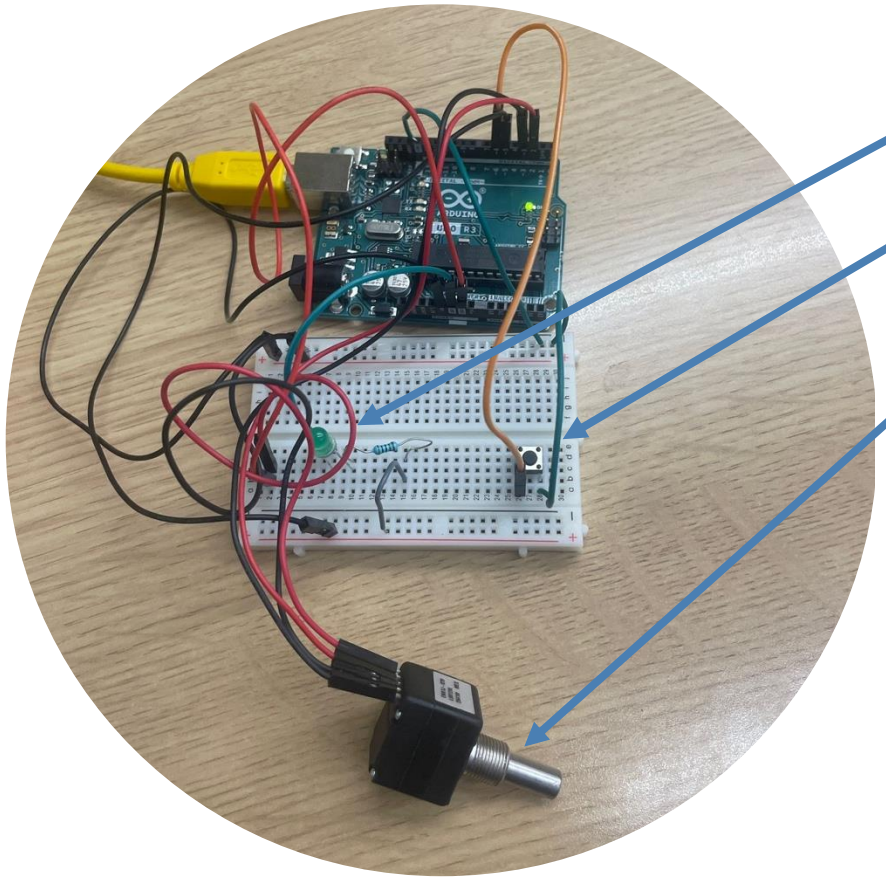


Pain and Safety

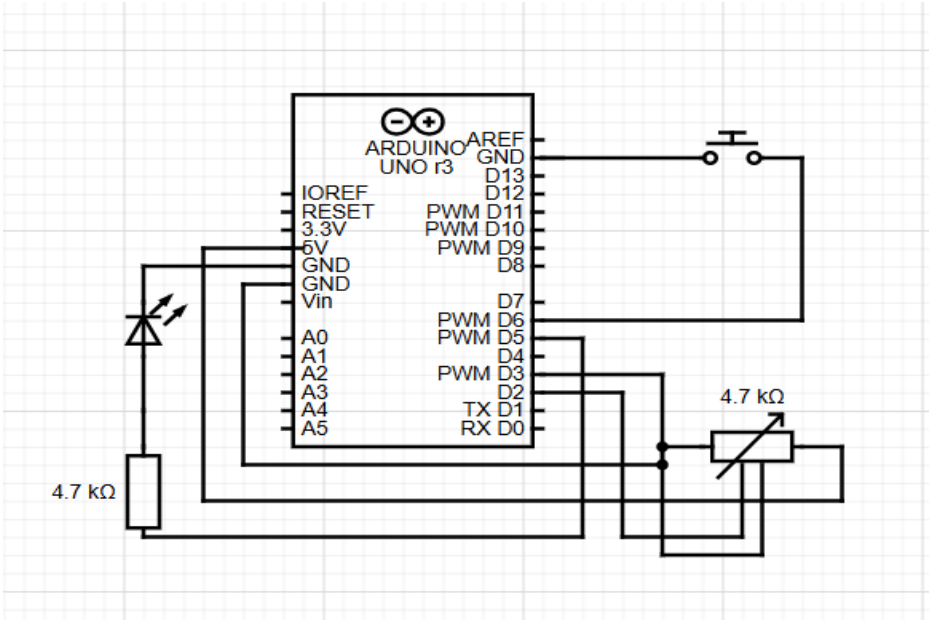
- **Polycentric hinge** - in line with knee biomechanics
- **Supportive functional knee brace** – distributes load
- **Ergonomic padded design** – prevents scraping skin
- **No internal battery** – limited fire risk
- **5v Arduino** – low temperatures, low risk of shocks



Internal Circuit



- **LED** – indicates when the device is active
- **Button** – starts and ends recording
- **Rotary Encoder** – captures angular data
- **Data Output** – synchronized angle and patella depth data



Software-User Interaction

Stage One – Preparation

Connect devices (iPhone, Arduino, USB Hub)



Stage Two – Clinical Marking

Mark points with dot (0º, 45º, 90º)

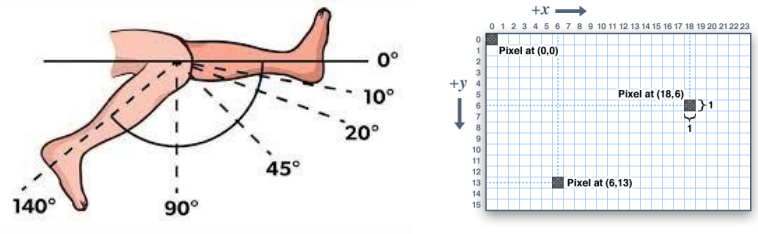
Input pen RBG colour values to script



Stage Three – Live Tracking

Initial dot coordinates recorded

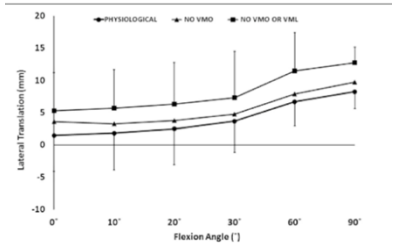
Depth and angle data recorded simultaneously



Stage Four – Calculation

Script calculates mediolateral shift for each frame

Data is exported to Excel



Conclusions

- Full Field of View – **9.1cm width** covers required 1.93cm
- Accuracy - **±1.013mm**
- Safety - Material safety factor of **10**; USB connection to eliminate battery risk
- Cleaning – Suitable for sanitisation with **96% ethanol**
- Standards - **Class 1m**, aligned with ISO and IEC
- Patient Perspective – Non-invasive design leading to quicker assessment and **faster pain relief**



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Live Demonstration

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Live Demonstration

